

Multiple Choice Quiz

Distributed Name Resolution

Total Questions: 10

Time Allowed: 20 minutes

Instructions: Choose the **best** answer for each question. Each question carries equal marks.

Question 1

What are the primary differences between flat names and structured names in terms of machine and human utility?

- A. Flat names are easy for humans to remember, while structured names are optimized for machines
 - B. Both flat and structured names offer the same level of human readability and machine efficiency
 - C. Flat names (e.g., cryptographic keys) are machine-friendly for verification but hard for humans to memorize, while structured names are hierarchical and human-readable
 - D. Structured names provide faster cryptographic verification than flat names
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Question 2

Identify the three node types found in a naming graph and explain their functions.

- A. Root Nodes (absolute starting points), Directory Nodes (junctions with mapping tables), and Leaf Nodes (actual entities or data)
 - B. Root Nodes, Branch Nodes, and Terminal Nodes
 - C. Directory Nodes, File Nodes, and Link Nodes only
 - D. Central Nodes, Edge Nodes, and Leaf Nodes with no specific functions
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Question 3

What is a “closure mechanism,” and why is it necessary for the resolution process?

- A. A mechanism that automatically closes all open network connections after resolution
- B. A security feature that encrypts the resolution path
- C. A backup system that restores previous name mappings
- D. The implicit starting knowledge (e.g., environment variables or superblock offsets) required to begin the resolution process

Question 4

How does a hard link differ from a symbolic link in a naming graph?

- A. A symbolic link points directly to the same node identifier, while a hard link stores a path string
 - B. A hard link allows multiple paths to point to the exact same node, while a symbolic link stores a path that forces re-resolution from another point**
 - C. Both hard links and symbolic links function identically in resolution
 - D. Symbolic links are more efficient because they never require re-resolution
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Question 5

What three elements must be resolved to merge a foreign name space into a local one through mounting?

- A. User credentials, file permissions, and network speed
 - B. Server IP address, username, and password only
 - C. Access Protocol (e.g., NFS), Server Name (often via DNS), and Mounting Point in the local graph**
 - D. File size, creation date, and owner information
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Question 6

Compare the scale, responsiveness, and replication requirements of the Global layer versus the Managerial layer.

- A. The Global layer operates worldwide with lazy responsiveness (seconds) and requires many replicas, while the Managerial layer operates at department scale with immediate responsiveness and no replicas**
 - B. Both layers have identical scale, responsiveness, and replication needs
 - C. The Managerial layer requires more replicas than the Global layer
 - D. The Global layer provides immediate responsiveness, while the Managerial layer is slower
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Question 7

In the context of name resolution algorithms, how does iterative resolution function?

- A. The resolver handles the entire query chain and returns only the final result to the client
- B. The client sends one query and waits passively for the answer
- C. All servers push updates to the client proactively**

D. The client's resolver queries a server, receives a referral, and sequentially follows referrals until the final address is found

Question 8

What is the primary advantage of recursive resolution regarding system efficiency?

- A. It places the maximum burden on the client to reduce server load
 - B. Intermediate servers learn and cache the hierarchy on the return path, enabling tree-wide learning and fewer future trips for clients
 - C. It completely eliminates the need for caching
 - D. It always requires more network round trips than iterative resolution
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Question 9

How does the Domain Name System (DNS) utilize a hybrid architecture to maintain global stability?

- A. DNS uses only recursive resolution at every level of the hierarchy
 - B. DNS uses only iterative resolution even at the edge for maximum decentralization
 - C. DNS uses strictly iterative resolution at the core (Root/TLD) to prevent overload, while allowing recursive resolution at the edge (ISP/Local) for better caching and speed
 - D. DNS avoids both iterative and recursive resolution entirely
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Question 10

What is the recommended safeguard for security when traversing global systems with potentially untrusted nodes?

- A. Validate the data itself using cryptographic signatures rather than trusting the path it traveled
 - B. Always trust the path if it comes from a well-known server
 - C. Use only iterative resolution to avoid malicious intermediate nodes
 - D. Disable all caching to prevent poisoned data from being stored
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Answer Key

Question	Correct Answer	Explanation
1	C	Flat names are machine-efficient (cryptographic keys) but hard for humans. Structured names are

Question	Correct Answer	Explanation
2	A	hierarchical and human-readable. Naming graphs contain Root Nodes , Directory Nodes , and Leaf Nodes with specific functions.
3	D	A closure mechanism provides the necessary starting knowledge (e.g., \$HOME or superblock) to begin resolution.
4	B	Hard links point to the same node ID. Symbolic links store a path and force re-resolution.
5	C	Mounting requires resolving the Protocol , Server Name , and Mount Point .
6	A	Global layer = worldwide, lazy, many replicas. Managerial layer = local, immediate, no replicas.
7	D	In iterative resolution, the client follows referrals step-by-step until the answer is found.
8	B	Recursive resolution allows intermediate servers to cache results on the return path, improving future efficiency.
9	C	DNS uses iterative resolution at the core (to avoid overload) and recursive resolution at the edge (for caching).

Question	Correct Answer	Explanation
10	A	Security is achieved by validating data via cryptographic signatures, not by trusting the path.

End of Quiz

Correct Answer Distribution: $A = 3$ / $B = 2$ / $C = 3$ / $D = 2$

This quiz tests understanding of distributed naming graphs, resolution algorithms, and architectural layers.